

Future languages

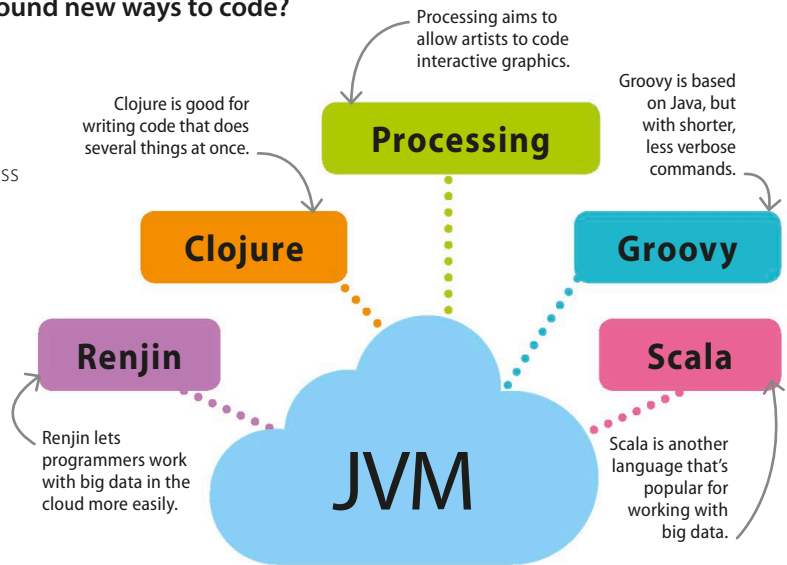
Computer-programming languages have come a long way in a relatively short time. In five years' time, will programmers still be using the languages of today or will they have found new ways to code?

Rising stars

A number of programming languages are rapidly increasing in popularity. R is designed for statistical programming and is useful for programs that process a lot of data. Go is very readable and good for networking. It's used by many large organizations. Haskell is a functional language that encourages better programming practices. Rust is based on C, but also includes elements of Haskell. TypeScript is a version of JavaScript with stricter rules, which results in safer code.

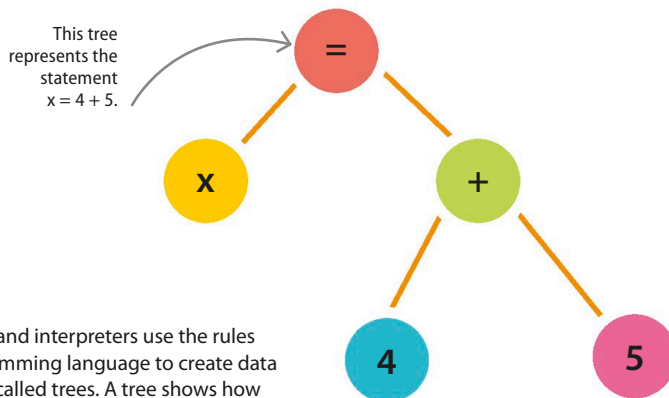
▷ Java Virtual Machine

Several emerging languages can be run on any computer that has a Java Virtual Machine (JVM) implementation. This is an advantage for the Internet of Things as the JVM can take inputs from devices that run different programming languages.



Creating a language

A number of things have to be considered when creating a new programming language, such as the styles of programming it allows, the existing language it will be written in, and whether the language will be compiled or interpreted. The next step is to create a grammar for the language. This is a set of rules defining how programs can be constructed. Once the grammar is defined, it can be used to write a compiler or interpreter for the language.



▷ Trees

Compilers and interpreters use the rules of a programming language to create data structures called trees. A tree shows how parts of a statement are connected.

IN DEPTH

Domain-specific languages



Programmers occasionally create domain-specific languages that are designed for writing programs to solve problems in one specialized area. Some examples include Verilog, used by hardware and computer chip designers, Logo, an early educational language that allowed children to move a turtle-shaped robot around the screen, and SQL, a language used for working with databases.